Table 2. MR and colocalization results for *TSHR*, *IGF1R* and *CTLA4* across five disease datasets

Gene	Dataset	IVs	OR (95% CI)	*P* value	Q-test *P*	PP.H2	PP.H3	PP.H4	PP.H4, p12 = 10^{-6}
TSHR	BBJ GD	1	0.12 (0.07–0.21)	1.09×10^{-14}*†	NA	<0.001	0.049	**0.951**	0.661
TSHR	FinnGen GD	1	0.08 (0.05–0.12)	3.19×10^{-32}*†	NA	<0.001	0.736	0.264	0.035
TSHR	UKB hyperthyroidism	1	0.09 (0.06–0.14)	8.77×10^{-28}*†	NA	<0.001	0.774	0.226	0.028
TSHR	FinnGen GO	1	0.10 (0.04–0.24)	2.82×10^{-7}*†	NA	<0.001	0.014	**0.986**	**0.875**
TSHR	FinnGen GO strict	1	0.15 (0.06–0.39)	9.02×10^{-5}*	NA	0.027	0.082	**0.891**	0.451
IGF1R	BBJ GD	4	1.56 (1.07–2.28)	0.0212*	0.902	0.690	0.236	0.073	0.008
IGF1R	FinnGen GD	3	1.07 (0.86–1.33)	0.539	0.882	0.731	0.259	0.010	0.001
IGF1R	UKB hyperthyroidism	4	1.35 (1.07–1.70)	0.0117*	0.250	0.400	0.196	0.404	0.063
IGF1R	FinnGen GO	3	1.41 (0.85–2.33)	0.182	0.309	0.623	0.346	0.032	0.003
IGF1R	FinnGen GO strict	3	1.52 (0.92–2.49)	0.0996	0.409	0.571	0.391	0.038	0.004
CTLA4	BBJ GD	1	0.18 (0.11–0.27)	5.45×10^{-15}*†	NA	<0.001	0.799	0.201	0.025
CTLA4	FinnGen GD	2	0.17 (0.08–0.36)	2.15×10^{-6}*†	0.0275	<0.001	0.746	0.254	0.033
CTLA4	UKB hyperthyroidism	2	0.21 (0.08–0.57)	0.00220*	0.00131	<0.001	0.047	**0.953**	0.672
CTLA4	FinnGen GO	2	0.17 (0.04–0.68)	0.0118*	0.0486	<0.001	0.022	**0.978**	**0.816**
CTLA4	FinnGen GO strict	2	0.17 (0.05–0.64)	0.00888*	0.0755	<0.001	0.021	**0.979**	**0.820**

MR estimates are ORs per approximately one SD higher genetically predicted blood expression: Wald ratio for one IV, multiplicative random-effects IVW for several. *TSHR* used one instrument, rs179252, in every dataset; its G allele raises blood expression by 0.105 SD per allele (F = 176), and its per-allele disease ORs were 0.77–0.82 (Table S3). The per-SD ORs therefore rescale a moderate per-allele association, and each single-IV *P* value equals the variant's disease-association z-test *P* value. *CTLA4* multi-IV estimates are concentrated in rs13030124, and three of the four have Q-test $P < 0.05$; in BBJ and UKB, two of four single-instrument omissions removed nominal *IGF1R* significance (Figure S2). * Nominal $P < 0.05$; † $P < 0.05/2{,}544$, the screen threshold, applied descriptively outside BBJ. PP.H2–H4 are colocalization posterior probabilities (H2, expression signal only; H3, distinct variants; H4, one shared variant) with p1 = p2 = 10^{-4} and p12 = 10^{-5} unless stated; bold, ≥0.80. They are not *P* values. BBJ colocalization pairs European eQTL with East Asian disease statistics. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. CI, confidence interval; eQTL, expression quantitative trait locus; IV, instrumental variable; IVW, inverse-variance weighted; MR, Mendelian randomization; NA, not applicable (one IV); OR, odds ratio; PP, posterior probability; SD, standard deviation; TED, thyroid eye disease; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.